Evidence-Qualified Alignment of Pathology Foundation Models with Morphologic, Molecular, and Outcome Axes in Prostate Cancer

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Abstract

Prostate-cancer AI can distinguish states, but clinicians need to know which human-interpretable quantitative axes its representations encode, whether those axes survive cohort and technical variation, and whether downstream decisions use them. We audited six non-interchangeable axes in frozen CONCH and Virchow representations: Grade/ISUP, tumor phenotype/content, PTEN loss, AR activity, SPOP mutation, and recurrence. For each, we assessed recoverability and, where possible, transport, conditioning, representation/sampling sensitivity, and endpoint stability; among the six axes, ISUP alone underwent a locked-head functional test. Grade and phenotype showed the strongest representation evidence and external transport. PTEN-related information was recoverable and AR activity showed positive pooled alignment, but independent or site evidence remained unresolved. SPOP was unsupported, whereas recurrence changed with endpoint, representation, and clinical reference. Removing an ISUP-correlated direction reduced two locked biochemical-recurrence heads' concordance beyond matched-random controls; refitted-head intervals included zero. This comparative map provides an actionable audit vocabulary: transported coordinates can anchor clinician review of agreement and disagreement, conditional or unsupported axes indicate where explanations should be withheld, and gaps prioritize external or functional validation. It enables clinician-understandable reliability audits and bounds future residual-signal discovery; it does not establish clinical benefit, indispensable use, external functional transport, or new biomarkers.

1 Introduction

Clinical adoption of artificial intelligence requires calibrated reliance rather than blind trust or a persuasive explanation attached to every output. Clinicians should be able to judge when an AI output has a defensible basis, when it should be challenged, and where the model's competence ends. Explanations that fit the decision context can support rational checking [3], but familiar explanations can also increase reliance when AI advice is wrong [4]. A clinically recognizable explanation is therefore not evidence of trustworthy model behavior unless the proposed link can itself be tested.

This is the motivating problem for prostate-cancer AI: beyond asking whether a model can separate clinically relevant states, we need to determine which human-interpretable quantitative axes it represents, whether each axis remains recoverable across cohorts and technical variation, and whether a downstream decision actually uses it. Answering these questions axis by axis can make model reliability inspectable in terms clinicians can contest, rather than reducing trustworthiness to predictive performance or a plausible visualization.

Clinically established quantitative and ordinal targets offer a possible interface for such testing. A pathologist can define, measure, contest, and communicate grade or tumor extent; molecular laboratories can provide assay-defined targets; and clinical records provide outcomes. If an AI signal is demonstrably linked to one of these targets, the proposed basis becomes inspectable: a pathologist can confirm that the tissue supports the named feature, reject the explanation when it does not, or identify a meaningful discordance. Concept-based methods provide precedents. TCAV tests prediction sensitivity to human-defined concepts, and concept bottleneck models make named concepts explicit and intervenable [5, 6]. Medical-imaging and pathology systems have likewise exposed confounders or connected influential regions and embeddings to expert-recognizable findings [7, 8, 9]. These studies also show why concept labels, embedding similarity, and actual decision dependence cannot be treated as interchangeable.

Three evidence stages are particularly easy to collapse. A target may be recoverable from a frozen representation; that relationship may reproduce across cohorts and technical settings; and a downstream disease head may functionally use the target. Only the third directly tests sensitivity of a particular prediction rule, and even faithful functional use would not prove clinical benefit. We therefore order the evidence as recoverability, reproducibility and qualification, functional-use testing, and finally prospective evaluation of clinician reliance and utility. Each stage enables, but does not replace, the next.

Clinical orientation and reference availability. The disease studied here is predominantly prostate adenocarcinoma. Routine hematoxylin-and- eosin (H&E) sections expose gland architecture, tumor amount, and tissue context but do not directly measure DNA alterations or transcriptional activity. The six study axes therefore span different biological levels [10, 18, 19]. Grade/ISUP measures architectural differentiation; tumor phenotype/content measures tumor presence or amount; PTEN loss and SPOP mutation are binary molecular alterations; AR activity is a continuous transcriptional pathway score. Recurrence is a patient outcome observed after treatment rather than a microscopic structure graded on the slide. They are not six interchangeable measures of "cancer severity."

Crucially, the slides were model inputs, not the only study data. Every reported alignment estimate paired a slide-derived frozen embedding with a target-specific reference at the same analysis unit. The encoder was not fine-tuned on these study labels; a lightweight probe used the paired references to test whether the embedding contained recoverable target information. Reference availability nevertheless differed: morphology had compatible external labels, molecular references were largely confined to TCGA-PRAD, and recurrence definitions were non-equivalent across resources. Missing reference data were treated as not evaluable, not as evidence of non-alignment. Unsupported required an available comparison that did not qualify the proposed signal, as for SPOP. The detailed target, reference, availability, metric, and cohort frames are provided in Supplementary Section S1.

Why the six prostate-cancer axes are not interchangeable. A positive result supports a different claim at each level. Grade and phenotype are direct morphology targets; PTEN, SPOP, and AR are indirect histology–molecular associations; and recurrence is shaped by biology, treatment, follow-up, censoring, and endpoint definition. Recovering a target shows that related information is accessible in the embedding, not that H&E replaces a molecular assay or that a later prediction head uses the target. Conversely, failure to recover or transport a target warns against invoking it as an explanation, but does not prove absence of the underlying biology.

Existing studies often summarize image–label association with one pooled correlation or area under the receiver operating characteristic curve. Such a number cannot distinguish target information from grade composition, institutional case mix, acquisition conditions, sampling, or endpoint definitions [12, 13, 14]. Nor does it show whether a fitted relationship survives a new cohort without refitting. We therefore define *representation–target alignment* as patient-level association or discrimination between a registered target and the out-of-fold or locked external output of a simple probe fitted to a frozen CONCH or Virchow representation [1, 2]. We qualify each target across recoverability, external transport, conditional increment, site behavior, representation and sampling sensitivity, and endpoint fidelity, retaining these as an evidence vector rather than a composite score.

Position of this study. This study compares Grade/ISUP, tumor phenotype/content, PTEN loss, SPOP mutation, AR activity, and recurrence to determine which targets are accessible in frozen pathology representations and how their evidence changes across cohorts, clinical adjustment, sites, representations, sampling, and endpoint definitions. Grade and phenotype provide the broadest test of external transport; PTEN, AR, and SPOP distinguish recoverable, conditional, and unsupported molecular states; and recurrence tests whether outcome-linked information retains its meaning across event definitions and comparators. Among these axes, Grade/ISUP has the most complete evidence chain because it is additionally tested in two locked biochemical-recurrence heads as a deliberately narrow internal functional-sensitivity extension. ISUP is thus the most developed functional example, not the sole organizing target. The study does not explain a deployed model, establish indispensable or externally replicated functional use, comprehensively test clinical increment, or measure clinician reliance or benefit.

Its primary task is comparative: to map the evidential status of all six axes without treating their labels, biological levels, or available validation opportunities as equivalent.

Contributions. This paper makes four contributions. First, it defines clinically interpretable targets as *contestable shared coordinates* through which a pathologist can agree or disagree with a clinically stated basis for an AI output. Second, it provides a comparative six-axis map across cohorts and two foundation models, distinguishing transported morphology, context-sensitive molecular and outcome associations, and unsupported evidence. Third, it turns the map into an actionable audit vocabulary: transported coordinates can anchor clinician review of agreement and disagreement; conditional or unsupported axes indicate where explanations should be withheld; and evidence gaps prioritize external or functional validation. Fourth, it provides a worked extension from representation availability toward functional testing by applying targeted erasure and variance-matched controls to an ISUP-correlated direction in locked CONCH- and Virchow-derived risk heads. This final experiment extends rather than organizes the six-axis comparison. The proposed audit vocabulary is intended for model evaluation, communication, and validation planning; whether showing it to clinicians improves decisions remains untested. Establishing this known-target map also defines the prerequisite for residual discovery: only after reliable shared coordinates and technical confounders are accounted for should remaining reproducible, decision-linked signals be investigated as candidate biomarkers rather than artifacts.

2 Results

The six targets occupy different biological and clinical levels: Grade/ISUP, tumor phenotype/content, PTEN loss, AR activity, SPOP mutation, and recurrence. The primary result is their comparative evidence hierarchy; ISUP is highlighted later only because it reached an additional functional-sensitivity experiment. Every numerical alignment result used an available target-specific reference paired to the slide-derived embedding at the registered analysis unit. Unequal reference coverage limited which external, conditional, or functional questions could be asked; it was not converted into a negative association. Thus, not evaluated or blocked denotes a missing prerequisite, whereas unsupported denotes an available comparison that did not qualify the proposed signal. Table 1 summarizes the headline states, and Supplementary Section S1 retains the detailed target, paired-reference, availability, metric, and biological definitions.

Table 1: Evidence-qualified candidate shared coordinates. States summarize the available axis-specific evidence: transportable means directional external transfer was retained; context-sensitive means interpretation narrowed under a clinical, site, representation, sampling, or endpoint audit; unsupported means the frozen primary design did not qualify the signal. Functional use refers specifically to intervention on a locked BCR head and is not implied by recoverability.

Target	State	Permitted interpretation	Functional use
Grade/ISUP	Transportable	Grade information is recoverable and directionally transports across compatible external resources	Internal exploratory locked-BCR-head sensitivity; indispensable, external, and clinical-increment use not established
Tumor phenotype/content	Transportable	Morphology-linked phenotype is recoverable and transports directionally; the external target is grade-derived tumor/benign status, not exact tumor-content truth	Currently blocked: same-cohort independent tumor-content truth unavailable
PTEN	Context-sensitive	PTEN-related information is recoverable within TCGA-PRAD but grade-independent increment and external transport are not established	Feasible follow-up; not run
AR activity	Context-sensitive	AR activity has positive pooled alignment but grade-independent increment and site transport remain unresolved	Feasible follow-up; not run
SPOP	Unsupported (frozen design)	Evaluated against an available genomic reference, but no reliable frozen-design signal was established	Not qualified: recoverability unsupported
Recurrence	Context-sensitive	Recurrence alignment is endpoint representation and covariate hierarchy conditioned	Not applicable as input erasure; prognostic validation required

2.1 The study separates recoverability, reproducibility, and functional use

We followed each target across the evidence axes permitted by its data: recoverability in a frozen representation, unchanged-probe external transport, and clinical, site, representation, sampling, or endpoint qualification (Figure 1). These are target-specific evidence states, not model-confidence scores or clinical thresholds. For grade alone, a separate internal analysis used locked BCR heads, target-specific erasure, and matched controls to test functional sensitivity. Phenotype was blocked by unavailable same-cohort independent truth; PTEN and AR were feasible but not run; SPOP lacked a qualified direction; and recurrence was the predicted outcome rather than an input target for erasure. External functional replication was not achieved.

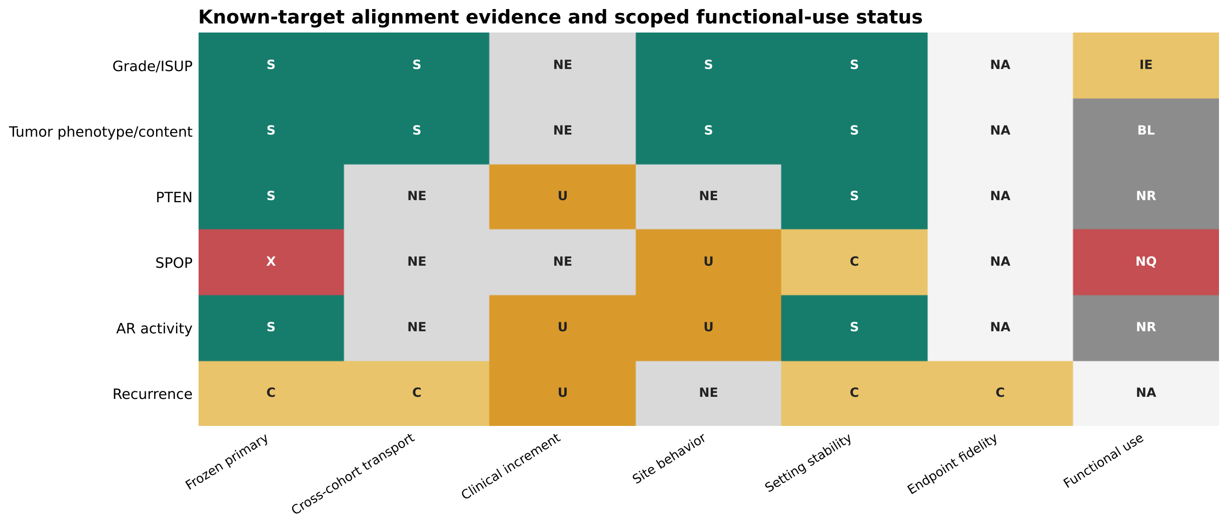

Figure 1: Evidence-qualified map of candidate shared interpretive coordinates. Rows are targets and columns are distinct evidence axes. Cell codes are S, supported; C, context-sensitive; U, unresolved; NE, not evaluated; NA, not applicable; and X, unsupported in the frozen design. In the functional-use column, IE denotes internal exploratory sensitivity for grade; BL, blocked because same-cohort independent phenotype truth was unavailable; NR, feasible but not run for PTEN and AR; NQ, not qualified because SPOP recoverability was unsupported; and NA, not applicable because recurrence is the predicted outcome. IE does not mean indispensable, externally reproduced, or clinically incremental use.

How to read the qualification map (Figure 1). Read rows from left to right. A supported cell applies only to the named axis and does not fill missing evidence elsewhere. The functional-use column distinguishes a completed internal test from an unavailable prerequisite, a feasible analysis not run, an unqualified direction, and a non-applicable question; probe recoverability must not be mistaken for head use.

2.2 Morphology-linked targets provide the best-qualified candidate coordinates

Grade and tumor phenotype were evaluated in NADT-Prostate and transferred without target-cohort refitting to PANDA; PRECISE supplied a separate session-level grade estimate. In NADT-Prostate, the patient-level CONCH grade probe had Spearman $\rho = 0.478$ (patient-bootstrap interval 0.170–0.712; 39 patients), and tumor phenotype had $\rho = 0.800$ (0.625–0.909; 39 patients). The locked grade probe retained positive associations in PANDA Karolinska ($\rho = 0.354$; 565 case images) and Radboud ($\rho = 0.519$; 572 case images), while the phenotype probe discriminated grade-derived tumor status with AUROCs of 0.786 and 0.871. A separate PRECISE evaluation yielded grade $\rho = 0.865$ across 17 imaging sessions (Figure 2).

These results support directional transport, not uniform effect sizes or functional use. PANDA phenotype used grade-derived tumor-versus-benign status rather than an independent same-patient quantitative tumor-content reference; it therefore shows directional phenotype transport, not exact external tumor-content agreement. Saved target-cohort intervals were unavailable for PANDA and PRECISE and were not reconstructed. The complete cohort, unit, denominator, metric, and uncertainty frame is in Supplementary Section S4.

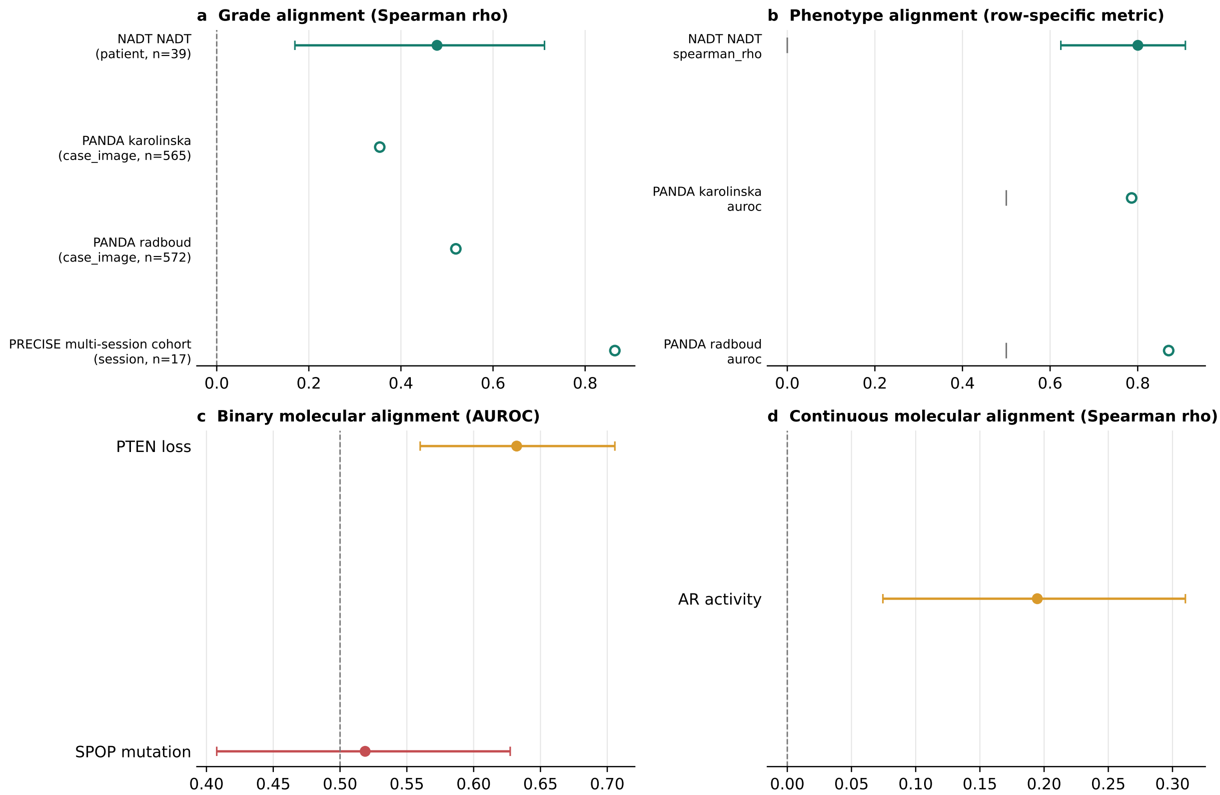

Figure 2: Recoverability and transport of known morphologic and molecular targets. **a**, Grade and phenotype estimates across source and external resources; metrics and analysis units are displayed rather than pooled. PANDA phenotype uses grade-derived tumor-versus-benign status. **b**, Frozen-primary molecular alignment in TCGA-PRAD; these are within-cohort recoverability results, not external transport. Whiskers are shown only where saved patient-bootstrap intervals were available.

2.3 Molecular coordinates separate into recoverable, conditional, and unsupported states

The molecular frame comprised 273 TCGA-PRAD patients. PTEN loss was recoverable with a frozen-primary CONCH AUROC of 0.632 (0.560–0.706), and AR activity had a positive Spearman correlation of 0.195 (0.074–0.310). SPOP was near chance (AUROC 0.519, 0.408–0.627), and its broader correlated seed-cell range crossed chance (0.348–0.679).

Conditional audits narrowed PTEN and AR (Figure 3). Held-out increments beyond grade were +0.019 AUROC (−0.025 to +0.068) for CONCH PTEN and +0.035 (−0.013 to +0.087) for Virchow PTEN. AR increments in R^2 were +0.004 (−0.036 to +0.042) and +0.019 (−0.015 to +0.051), respectively. The pooled AR direction was positive, but all six stored site-specific intervals crossed zero; evaluable SPOP site intervals included or touched chance, and one site had no positive event.

Recoverable molecular information was therefore not shown to add held-out information beyond grade or to transport across sites. Site remained entangled with acquisition and case mix. The SPOP genomic reference was available and was used, so its unsupported state is an evaluated lack of qualification in this frozen design, not a missing-label result. None of these findings establishes molecular diagnosis from H&E or use by a disease-prediction head.

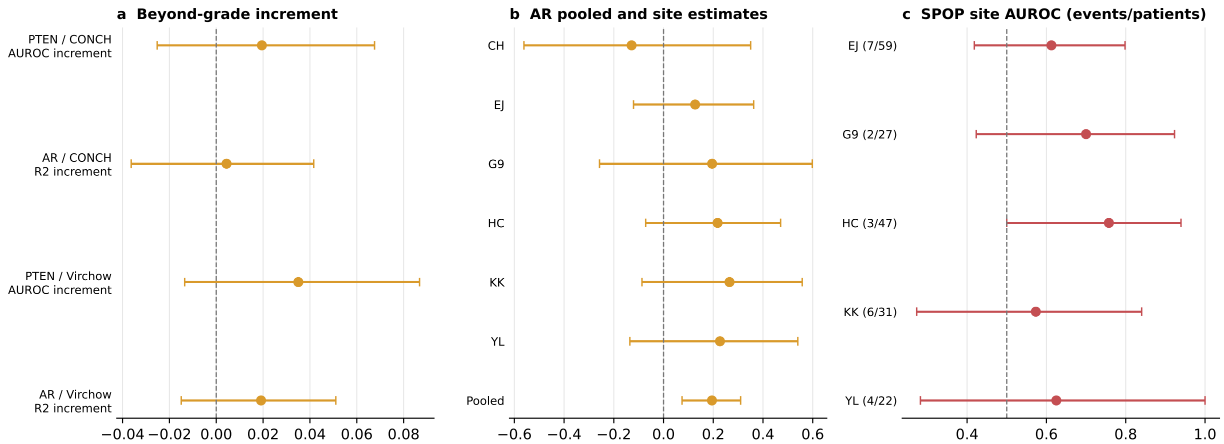

Figure 3: Conditional and site-specific narrowing of molecular alignment. **a**, Held-out PTEN and AR increments beyond grade for CONCH and Virchow; positive values indicate improvement after adding the image-derived score. **b**, Pooled and site-specific AR correlations. **c**, SPOP site-specific AUROCs with event and patient counts. Intervals crossing or touching the relevant null indicate unresolved evidence, not absence of biological signal.

2.4 Outcome alignment changes with endpoint and clinical reference

A post-hoc recurrence score fitted in LEOPARD was applied unchanged to TCGA-PRAD. In the 270-patient transfer frame, reconstructed with-tumor recurrence yielded a C-index of 0.673 (0.587–0.759; 57 events), whereas official TCGA-CDR PFI yielded 0.586 (0.482–0.689; 42 events). On 153 complete cases, adding the image score to grade changed C-index by +0.083 (+0.012 to +0.157; 30 reconstructed events) but had an integrated-Brier-score interval including zero. For official PFI, the C-index increment was +0.032 (−0.040 to +0.126; 15 events), and all paired C-index and integrated-Brier-score intervals included zero. Fuller clinical and site comparisons did not establish robust independent increment.

Recurrence is therefore endpoint- and reference-conditioned, not a validated prognostic biomarker. Concordance measures risk ordering rather than calibrated individual risk or clinical usefulness, and the exploratory PCA sweep was not an unbiased model-selection procedure. Supplementary Section S9 retains the endpoint definitions, complete paired comparison family, and outcome-contrast figure.

2.5 Representation audits bound sensitivity without selecting a universal model

Six targets were evaluated across 12 encoder–scale–tile configurations and five sampling seeds, yielding 360 correlated seed cells and 390 seed-matched contrasts. Grade, phenotype, PTEN, and AR retained their direction across observed configuration ranges; null-straddling occurred in 8 of 12 for SPOP and 1 of 12 for recurrence (Figure 4). Because these repeats reused patients, folds, and representations, they assess analytic sensitivity rather than independent replication or a universal CONCH-versus-Virchow ranking. Supplementary Section S8 contains the complete grid and the seed-matched encoder, physical-scale, and tile-budget contrast figure.

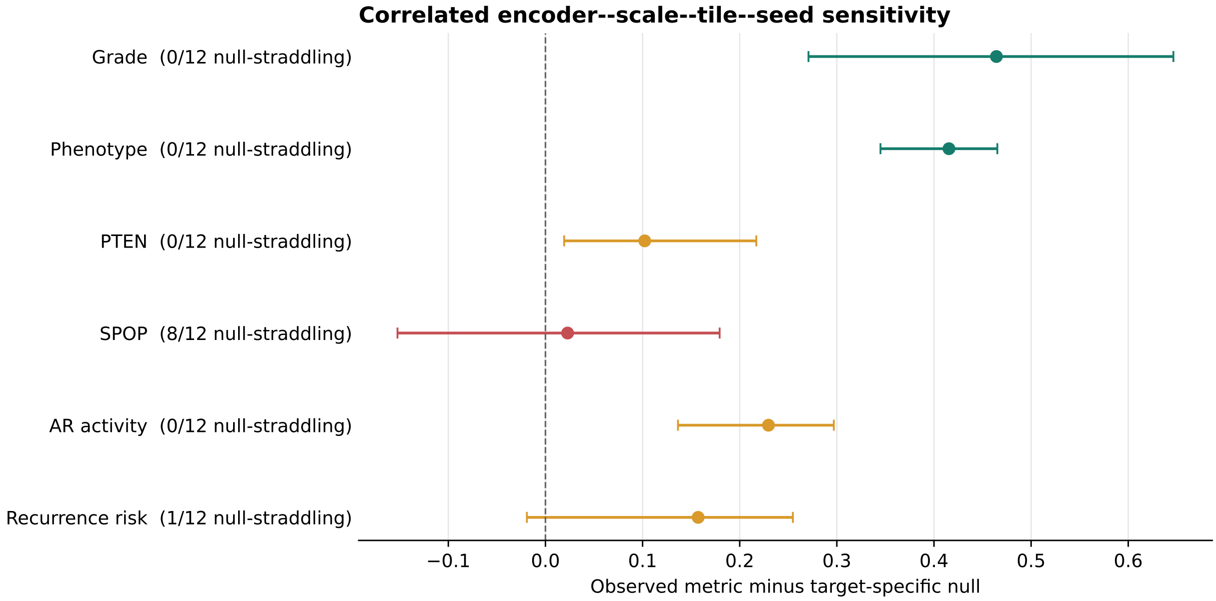

Figure 4: Target-specific representation and sampling sensitivity. Observed five-seed ranges across 12 encoder–scale–tile configurations are shown relative to each target's metric null. The 360 cells reuse cohorts and folds and therefore describe correlated internal sensitivity settings rather than independent validation.

2.6 ISUP provides the most developed extension from representation alignment to internal functional sensitivity

Grade/ISUP was the only axis for which the current data and protocol supported a locked-head extension. The exploratory frame comprised 392 TCGA-PRAD patients with 80 BCR events. CONCH and Virchow remained frozen; patient-disjoint ridge probes learned ISUP-correlated directions, and ridge-penalized Cox heads learned BCR risk. The heads were then held fixed while the ISUP-correlated direction was removed, with variance-matched random directions as controls.

ISUP was recoverable out of fold (Spearman $\rho = 0.615$ for CONCH and 0.658 for Virchow), and the full BCR heads had C-indices of 0.627 (0.553–0.694) and 0.632 (0.564–0.691). Targeted removal reduced concordance by +0.041 (0.011–0.073) and +0.023 (0.007–0.040), respectively, exceeding matched-random controls ($p = 0.0099$ for each). After refitting, differences narrowed to +0.014 (−0.006–0.035) and +0.010 (−0.003–0.022). Limited ISUP-plus-AI versus ISUP-only differences were −0.000 (−0.039–0.039) and −0.032 (−0.088–0.025). Intervals including zero indicate potential compensation and do not establish indispensable use or absence of clinical increment.

A protocol-locked clean rerun preserved all estimates and gate states; Supplementary Section S5A reports the 20/20 hash audit and detector boundary. This establishes computational reproducibility, not an independent-cohort replication. Independent-domain tumor-detector sensitivity remained unresolved, so the functional result remains whole-tissue and internal.

Grade therefore receives an *internal exploratory functional-sensitivity* state. It is the most developed example in the comparative map, but it is not promoted to indispensable, external, mechanistic, or clinically incremental use. No residual or unknown representation direction was evaluated as a new marker.

2.7 The final map distinguishes supported, conditional, and unsuitable coordinates

The combined hierarchy retains unlike metrics and missing evidence rather than averaging them. Grade and phenotype are the best-qualified representation coordinates; PTEN and AR are conditional; SPOP is unsuitable in the frozen design; and recurrence is endpoint-sensitive. Grade alone additionally has internal exploratory functional-sensitivity evidence.

Figure 5 links each clinical coordinate to its operational probe-defined feature and the available evidence about downstream judgment. This does not mean that only ISUP describes prostate cancer: phenotype is strongly represented, PTEN and AR are feasible follow-up tests that were not run, SPOP did not yield a qualified representation feature, and recurrence requires external prognostic validation. What is unique to ISUP is evidence that the tested BCR heads were sensitive to its correlated embedding direction.

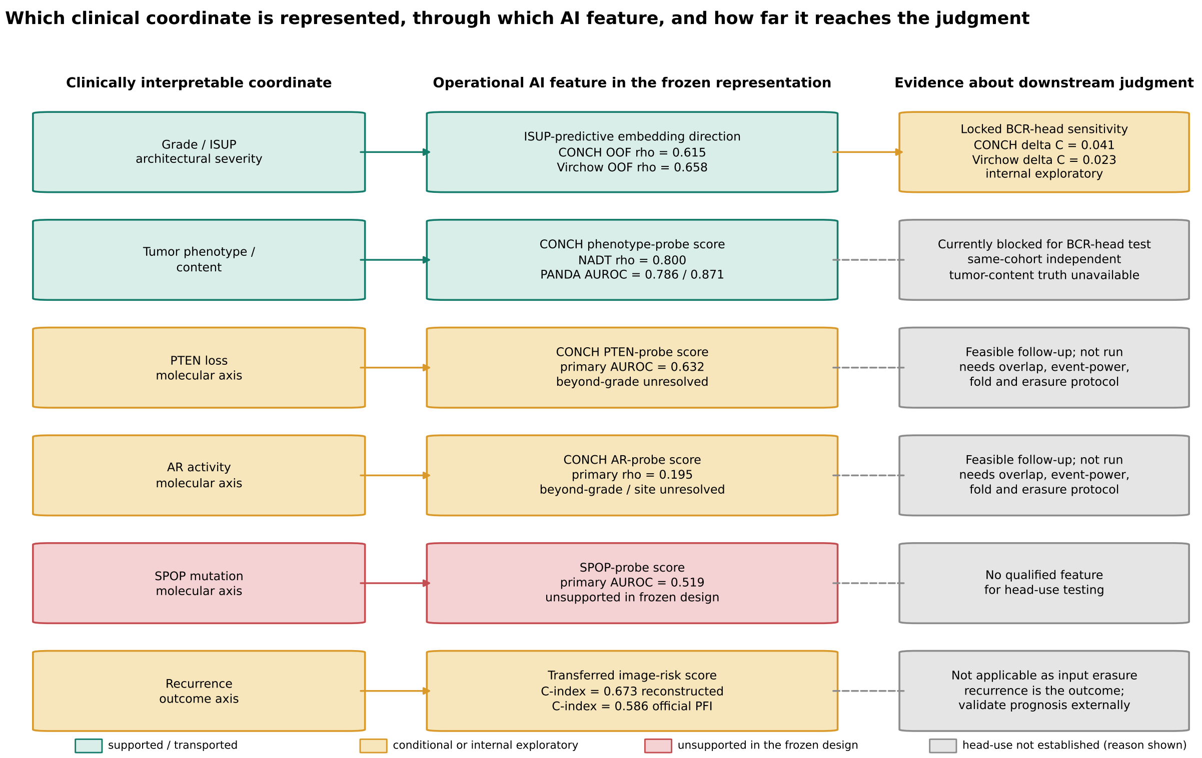

Figure 5: Human quantitative coordinate–AI feature–judgment linkage map. The middle column identifies the operational probe direction or score, not a localized gland, cell, or latent unit with independently established biological meaning. Solid arrows denote an evaluated link; dashed links end at representation evidence with the row-specific reason shown. Only ISUP has internal locked-BCR-head erasure evidence, repeated in CONCH and Virchow. The displayed coordinates do not completely explain an AI decision.

3 Discussion

The central result is not that one prostate-cancer target explains a foundation model. Six clinically interpretable axes occupied distinct positions in the evidence map: Grade/ISUP and tumor phenotype/content had the broadest representation-level transport; PTEN-related information was recoverable and AR activity aligned positively, but both became conditional; SPOP did not yield a qualified direction; and recurrence changed with endpoint, comparator, representation, and scale. The ISUP intervention extended one eligible axis from representation availability to internal functional sensitivity, but it does not make ISUP the sole organizing target.

This hierarchy matters because it turns familiar targets into contestable coordinates rather than post-hoc labels attached to an AI output. Transported coordinates can anchor clinician review of agreement and disagreement; conditional or unsupported coordinates indicate where an explanation should be withheld; and missing evidence identifies the next cohort, intervention, or endpoint to test. This supports reliability auditing in clinically intelligible terms, but whether presenting the map improves calibrated reliance or decisions remains a prospective question.

Reference availability constrains the hierarchy. The study did not ask a slide-only model to agree with six unobserved quantities: each estimate used an available paired reference. Grade and phenotype had the widest compatible external coverage, molecular references were largely confined to TCGA-PRAD, and recurrence definitions were non-equivalent. Missing external or same-patient functional truth therefore produced not-evaluated or blocked states, whereas SPOP was unsupported after comparison with an available genomic reference. The hierarchy reflects both biological accessibility in H&E and unequal opportunities for transport and functional validation.

The target-specific results reinforce this distinction. Morphology-linked signals transported most clearly, although PANDA phenotype used grade-derived tumor status rather than exact tumor fraction. PTEN and AR support conditional histology–molecular associations, not molecular diagnosis from H&E or BCR-head use. Recurrence carried endpoint-associated risk information, but its variation across event definitions and clinical references precludes a validated prognostic-biomarker claim. The correlated setting grid exposed analytic sensitivity without creating new patients, institutions, or independent replications.

Our approach also differs from methods that directly connect concepts to a trained prediction. TCAV estimates prediction sensitivity along a concept direction, and concept bottleneck models route predictions through explicit concepts [5, 6]. Related medical-imaging and pathology systems connect influential regions or embeddings to recognizable findings [8, 9]. Here, the primary contribution is comparative qualification of pre-existing targets in frozen representations. Targeted erasure is a secondary worked example, not a complete concept-grounded architecture or mechanistic-fidelity result.

For ISUP, fixed-head concordance fell more after targeted than matched-random removal in both encoders. Refit intervals nevertheless included zero, and the limited ISUP-only comparison was not a comprehensive clinical-increment study. The warranted claim is sensitivity of two internal BCR heads to an ISUP-correlated subspace, not necessary use of a human-equivalent grading mechanism or external functional transport. Exact clean-rerun agreement strengthens computational reproducibility but cannot upgrade the claim to external functional replication. Absence of a functional-use result is therefore not one common negative result: the other axes stopped because of missing same-patient truth, unrun but feasible protocols, an unqualified direction, or a non-applicable intervention question.

Several limitations remain. Stronger morphology transport partly reflects greater external reference availability. Some external rows lacked retained intervals or event accounting; site remained confounded with acquisition and case mix; and recurrence analyses were post hoc or limited by endpoint differences and modest event counts. Independent-domain detector sensitivity remained below its gate, so ISUP functional evidence is whole-tissue and internal. Calibration, treatment utility, autonomous diagnosis, population-wide generalization, clinician trust, reliance, diagnostic accuracy, and patient outcomes were not evaluated. Familiar explanations can themselves increase inappropriate reliance [4], making prospective testing essential.

Finally, the map creates a disciplined discovery space rather than reporting a new biomarker. After reliable known coordinates and technical confounders are jointly accounted for, a residual signal should advance only if it recurs across models and independent cohorts, contributes to a locked prediction head, can be defined as a reproducible human-measurable feature, and receives biological or clinical validation. This separates potential biomarker discovery from artifacts while preserving the present contribution: a qualified map of what is shared, conditional, unsupported, and still untested.

4 Conclusion

The study was motivated by the need to make prostate-cancer AI reliability clinically interrogable: which quantitative axes are represented, which remain stable across cohort and technical changes, and which are used by a downstream decision. The comparative hierarchy answers the first two questions across six non-interchangeable axes. Grade/ISUP and tumor phenotype/content were the strongest transported representation coordinates; PTEN loss and AR activity were conditional; SPOP mutation was unsupported in the frozen design; and recurrence was endpoint- and reference-sensitive. The third question advanced only to a scoped internal ISUP functional-sensitivity example and did not establish indispensable or externally transported head use.

This qualified map provides contestable coordinates for model audit. Transported coordinates can anchor clinician review of agreement and disagreement, conditional or unsupported axes show where explanations should be withheld, and evidence gaps prioritize external or functional validation. Unavailable labels are not negative results. These uses do not demonstrate pathologist agreement, calibrated reliance, clinical benefit, or complete explanation of AI representations or decisions.

The map does not itself demonstrate a new residual marker; it defines a disciplined starting point for potential new-biomarker discovery. After jointly accounting for known clinical–pathology targets and technical confounders, residual decision-linked signals must recur across models and independent cohorts, contribute to a locked prediction head, become reproducible human-measurable features, and receive biological or clinical validation before being considered candidate biomarkers. External functional validation, clinician studies, and residual-marker discovery therefore remain distinct follow-up studies.

5 Methods

The Methods retain the information needed to interpret and reproduce the main claims; implementation details, complete sensitivity families, and missingness accounting are provided in Supplementary Sections S1–S12. The analyses, endpoints, folds, and frozen estimates were inherited from the source prostate-biomarker study without retrospective optimization for this manuscript.

5.1 Study design and interpretation contract

Representation–target alignment was defined as association or discrimination between a registered target and the patient-disjoint out-of-fold or locked external output of a probe fitted to a frozen representation. The primary design compared Grade/ISUP, tumor phenotype/content, PTEN loss, SPOP mutation, AR activity, and recurrence under target-appropriate metrics and qualification axes. It tests whether target-related information is recoverable and reproducible, not whether a disease-prediction head uses the same human concept. A secondary extension tested Grade/ISUP with locked BCR heads, targeted erasure, and matched controls because this axis had the required recoverable direction and available internal outcome frame. This unequal testing depth is reported as part of the six-axis evidence map rather than interpreted as evidence that grade is uniquely important. Neither analysis establishes mechanistic equivalence, indispensable use, complete explanation, clinical utility, or an observed effect on clinician trust or reliance. Clinician trust, reliance, diagnostic performance, and clinical utility were not study endpoints.

Targets were evaluated on the evidence axes permitted by their data: primary recoverability, cross-cohort transport, conditional increment, site behavior, representation and sampling sensitivity, and endpoint fidelity. The resulting states—transportable, context-sensitive, unsupported in the frozen design, unresolved, not evaluated, or not applicable—are evidence summaries rather than model-confidence or clinical-action scores.

5.2 Cohorts, targets, and analysis units

NADT-Prostate supplied source grade and tumor-phenotype labels; PANDA supplied Karolinska and Radboud case images and official ISUP grades for locked transport; PRECISE supplied session-level pathologist Gleason scores; TCGA-PRAD supplied histology, grade, PTEN loss, SPOP mutation, AR activity, and recurrence-style endpoints; and LEOPARD supplied BCR labels for fitting the transferred recurrence score [11, 10, 17, 15, 16]. Only evaluable target rows were used, and missing or uncertain outcomes were not recoded as negative. Patient, case-image, session, and slide effects remained explicitly distinct.

For NADT, predictions and labels were aggregated by patient and phenotype was the mean tumor fraction. PANDA sampling capped each institution–ISUP stratum at 100 cases before tissue filtering; its phenotype target was ISUP 0 versus ISUP at least 1. Recurrence-style definitions were not pooled or renamed: reconstructed with-tumor recurrence, strict recurrence, TCGA-CDR PFS and DFS, and official PFI retained separate identities [19]. Full cohort, target, event, and analysis-unit definitions are given in Supplementary Sections S1 and S9.

5.3 Frozen representations, probes, and evaluation

CONCH and Virchow encoder weights remained frozen. Tissue-qualified tiles were converted to embedding vectors and averaged into slide representations; patient endpoints then used patient aggregation. The sensitivity program crossed encoder, native or shared physical scale, tile budgets of 16, 32, or 64, and five sampling seeds on fixed cohorts and folds.

Continuous targets used standardized ridge regression and binary TCGA-PRAD targets used standardized logistic regression. The fitted coefficients define a target-predictive direction rather than a localized gland or cell type. Five patient-disjoint folds produced out-of-fold scores; locked transfer applied the source probe and preprocessing without target-cohort refitting.

Metrics followed target type: Spearman correlation for continuous or ordinal targets, AUROC for binary targets, and C-index for censored survival risk. AUROC is threshold-free and does not choose a clinical sensitivity/specificity threshold; a C-index is not a calibrated survival probability. The saved discovery family used two-sided Spearman or Mann–Whitney tests with Benjamini–Hochberg false-discovery-rate control, and conditional audits used a grade-stratified permutation test. Supplementary Sections S2, S3, S6, and S7 retain full specifications, nulls, multiplicity, resampling, and statistical definitions.

5.4 Conditional, setting, and outcome qualification

PTEN and AR conditional analyses compared clinical-only, image-only, and combined models in nested patient-disjoint folds. Held-out increments were change in AUROC for PTEN and change in R^2—the proportion of held-out variation explained—for AR. AR site effects used slide-level Spearman correlation with patients resampled as clusters; site was not interpreted causally. Encoder, scale, tile-budget, and seed repeats reused patients and folds, so their 390 matched contrasts describe analytic sensitivity rather than independent replication (Supplementary Section S8).

The post-hoc recurrence score was fitted with a Cox model in LEOPARD and applied unchanged to TCGA-PRAD. Reconstructed recurrence and official PFI were evaluated separately. Paired comparisons reused the same complete cases and bootstrap draws; positive C-index or integrated Brier-score deltas favored augmentation under the registered definitions. The PCA sweep was exploratory. Supplementary Section S9 provides endpoint definitions, delta directions, and all paired comparisons.

5.5 Secondary ISUP functional-sensitivity extension

The FM6 extension used 27,968 whole-tissue tiles aggregated to 392 TCGA-PRAD patients with 80 BCR events. Within five outer patient-disjoint folds, 64 principal components summarized each frozen representation, ridge regression learned an ISUP direction, and a penalized Cox model learned BCR risk. The primary intervention locked the Cox coefficients while projecting the normalized ISUP direction from held-out representations. One hundred variance-matched random directions and 20 label-permuted directions served as controls; secondary analyses refitted the head after erasure and compared ISUP plus AI with ISUP alone. Patient-bootstrap intervals used 2,000 resamples.

The analysis remained whole-tissue because the detector failed its sensitivity gate in both held-out PANDA providers. No equivalent intervention was performed for the other axes. A protocol-locked rerun required unchanged estimates, gates, and 20 nonvolatile hashes. The gate labels were R, recoverability; A, internal BCR association; U, functional sensitivity; and T, equivalent external transport. R, A, and U passed at the internal exploratory level; T was not tested. Supplementary Section S5A retains the complete intervention, detector, rerun, and claim ceiling.

5.6 Uncertainty, missingness, and provenance

Where supported by the saved source, estimates used 95% percentile patient-bootstrap intervals; the patient bootstrap resamples patients rather than slides. AR site intervals used patient-cluster resampling. Unavailable intervals, event counts, single-class sites, undefined replicates, and legacy bootstrap accounting were preserved rather than reconstructed. An interval containing the registered null was treated as unresolved evidence, not proof that the biological relationship is absent; an interval excluding the null did not substitute for transport, functional-use, or clinical-utility evidence. The Benjamini–Hochberg procedure controlled the false-discovery rate across the saved discovery family, whereas target-specific bootstrap intervals described estimation uncertainty. No multiple imputation was performed for the paired recurrence analysis. Supplementary Section S10 provides the complete accounting.

Analyses used the repository's locked Python environment, fixed seeds, and auditable entry scripts. The builder verifies owner, byte count, and SHA-256 hash for every source registered in `provenance/source_evidence_manifest.csv`; figures and tables are regenerated from saved source tables, and a mismatch fails the build. Claim, endpoint, and numeric registries provide row-level lineage. The prostate-biomarker project remains owner of the frozen source analyses, while QFM owns this derivative manuscript.

Large language model assistance. OpenAI Codex assisted with code editing, test authoring, figure-code preparation, and language revision. The authors reviewed and verified all AI-assisted outputs and take full responsibility for the final manuscript.

5.7 Ethics approval and consent to participate

This work was a secondary analysis of deidentified data obtained from previously published public or access-governed repositories. No new participants were recruited, no intervention or specimen collection was performed, and the authors had no access to direct identifiers. The original studies obtained ethical approvals and informed consent or waivers as reported by the source publications and repositories. No additional ethical approval or participant consent was required for this secondary analysis. All data were used in accordance with the applicable repository access and data-use requirements. Consent for publication was not applicable because the manuscript reports aggregate results and contains no identifiable participant information.

Data Availability

The public cohorts used in this study retain their source-specific access and reuse terms. Digital Object Identifiers (DOIs) provide the persistent repository links below. NADT-Prostate is available from The Cancer Imaging Archive at `https://doi.org/10.7937/TCIA.JHQD-FR46` [11]; PANDA data are distributed through the PANDA challenge under its platform terms [10]; TCGA-PRAD histology and molecular data are available through the Genomic Data Commons, with the referenced TCGA clinical endpoint fields available through cBioPortal; and PRECISE is available from Zenodo at `https://doi.org/10.5281/zenodo.20721779` [17]. LEOPARD is an access-governed challenge resource: its training data and outcome labels remain subject to the LEOPARD challenge and dataset terms, and its public challenge report is cited here [15]. Publication-facing aggregate numerical source tables used to generate the article's figures and tables are supplied with the public code repository. Patient-level derived analysis artifacts and local provenance manifests are not redistributed.

Code Availability

Analysis-display code, aggregate source tables, provenance registries, source-analysis snapshots, and figure-reproduction tests are available in the public release repository under the annotated tag `v1.0.5-submission`. The auditable entry point is `build_publication_artifacts.py`; `REPRODUCIBILITY.md` identifies the original analysis entry points and resource requirements. The builder verifies source hashes and mapped headline values before regenerating all publication figures. Reference copies of the submitted PDFs are included; editable manuscript source is maintained by the authors. No source model was retrained or refitted solely for this interpretive re-analysis.

Repository-authored software and documentation are released under the MIT License. Dataset copies, whole-slide images, cached embeddings, pretrained model weights, access-governed LEOPARD artifacts and patient-level derived outputs are not redistributed; they must be obtained or reconstructed from their original sources under the applicable access terms and are not covered by the repository's software license.

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Author Contributions

J.H.K. had overall responsibility for the manuscript, conceived and designed the study, conducted the experiments, and wrote the manuscript. D.H.S. contributed to the study concept and its verification, assessed the medical validity of the study, and reviewed the manuscript. H.C. reviewed the manuscript and curated the data. I.L. assessed the overall scientific integrity of the manuscript.

Additional Information

Competing Interests

The authors declare no competing interests.

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